A Category-Large Class of Norm-Weak USC Points in c_0^{**}

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Source Question

In *Computing sub differential limits of operators on Banach spaces*, arXiv:2212.09650, Rao studies norm-weak upper semicontinuity of the preduality map. After Proposition 9, the paper asks how large, in the Baire category sense, the set of norm-weak usc points can be on X^{**} when X is an M-ideal in its bidual.

Result

We give a positive category answer for the prototypical non-reflexive M-embedded space $X = c_0$.

Theorem 1. *Let $X = c_0$, so that $X^{**} = \ell_\infty$ and $X^{***} = ba(\mathbb{N})$. The set of points of norm-weak upper semicontinuity for the preduality map on X^{**} contains a dense open subset of ℓ_∞. In particular it is residual.*

Idea of the Proof

For $a \in \ell_\infty$, norm-weak usc is equivalent, by the criterion quoted in the source paper from Godefroy–Indumathi, to weak-star density of the predual state space $S_a \subset \ell_1$ in the full state space $S^a \subset ba(\mathbb{N})$.

The dense open set is made of sequences whose norm is attained with a strict gap from the remaining coordinates. For such a sequence, all norming finitely additive measures live on the Stone–Cech closures of the maximum-coordinate sets. Atomic measures on the coordinates are weak-star dense in all measures on those closures. This gives the required state-space density.

Proof

We work over the real scalars. Identify ℓ_∞ with $C(\beta\mathbb{N})$. For $a = (a_n) \in \ell_\infty$, write $\widehat{a}$ for its continuous extension to $\beta\mathbb{N}$, and put $M = \|a\|_\infty$.

Let $\mathcal{G}$ be the set of all nonzero $a \in \ell_\infty$ such that, for

$$E_+ = \{n : a_n = M\}, \qquad E_- = \{n : a_n = -M\}, \qquad E = E_+ \cup E_-,$$

one has $E \neq \varnothing$ and

$$\sup_{n \notin E} |a_n| < M.$$

Lemma 1. *Every $a \in \mathcal{G}$ is a point of norm-weak usc for the preduality map on $\ell_\infty = c_0^{**}$.*

Proof. Let $a \in \mathcal{G}$. Choose $\eta > 0$ such that $|a_n| \le M - \eta$ for all $n \notin E$. By continuity of $\widehat{a}$, the subset of $\beta\mathbb{N}$ on which $\widehat{a} = M$ is exactly $\overline{E_+}^{\beta\mathbb{N}}$, and the subset on which $\widehat{a} = -M$ is exactly $\overline{E_-}^{\beta\mathbb{N}}$.

The predual state space $S_a \subset \ell_1$ consists precisely of the signed atomic measures u supported on E, positive on E_+, negative on E_-, and with total variation 1. The full state space

$$S^a = \{\mu \in ba(\mathbb{N}) : \|\mu\| = 1,\ \mu(a) = M\}$$

consists of signed regular measures on $\beta\mathbb{N}$ whose positive variation is supported on $\overline{E_+}^{\beta\mathbb{N}}$ and whose negative variation is supported on $\overline{E_-}^{\beta\mathbb{N}}$.

Atomic probability measures on a dense subset of a compact Hausdorff space are weak-star dense in the probability measures on that compact space. Applying this separately on $\overline{E_+}^{\beta\mathbb{N}}$ and $\overline{E_-}^{\beta\mathbb{N}}$, with the Jordan decomposition of μ, shows that every $\mu \in S^a$ is a weak-star limit of elements of S_a. Thus S_a is weak-star dense in S^a, which is equivalent to norm-weak usc by the source criterion. $\square$

Lemma 2. *$\mathcal{G}$ is open and dense in ℓ_∞.*

Proof. Openness follows from the strict gap. If $a \in \mathcal{G}$ and $\sup_{n \notin E} |a_n| \le M - \eta$, then every sufficiently small ℓ_∞-perturbation still has its norm attained on the old maximum set and remains separated from the remaining coordinates.

For density, fix $a \in \ell_\infty$ and $\varepsilon > 0$. Let $M = \|a\|_\infty$. Choose n with $|a_n| > M - \varepsilon/3$. Pick δ with $M - |a_n| < \delta < \varepsilon$, and define

$$b = a + \delta \operatorname{sgn}(a_n) e_n,$$

using either sign if $a_n = 0$. Then $\|b - a\|_\infty = \delta < \varepsilon$, and the n-th coordinate of b has absolute value $|a_n| + \delta > M$, while all other coordinates have absolute value at most M. Hence $b \in \mathcal{G}$. $\square$

The theorem follows immediately from the two lemmas. Since the norm-weak usc points contain the dense open set $\mathcal{G}$, their complement is contained in the nowhere dense set $\ell_\infty \setminus \mathcal{G}$.

Verification and Limitations

The proof uses only the source paper's state-space criterion and the standard identification $\ell_\infty = C(\beta\mathbb{N})$, $ba(\mathbb{N}) = C(\beta\mathbb{N})^*$. It does not address arbitrary M-embedded spaces, nor the operator-space intermediate question in Remark 13 of the source paper.

Novelty Check

Local index search found no exact packet for arXiv:2212.09650. Web searches for the phrases `norm-weak usc c0 preduality map`, `norm-to-weak upper semi-continuity duality c0`, and close variants did not locate an existing statement of this c_0 category subcase.

Source Evidence

when is $lim_{t\to 0+}\frac{\|x+t\tau\|-\|x\|}{t} = sup\{\tau(f) : f \in S_x\}$ for all $\tau \in X^{**}$? We know that $lim_{t\to 0+}\frac{\|x+t\tau\|-\|x\|}{t} = sup\{\tau(f) : f \in S^x\}$ for all $\tau \in X^{**}$.

Thus we need equality of the suprema for all $\tau \in X^{**}$. This is clearly seen to be equivalent to S_x being weak*-dense in S^x.

We next recall that for $x^* \in X^*$ that attains its norm at a $x \in X_1$, the map $\rho(x^*) = S_x$ is called the preduality map. We note that points of X^* that attain the norm, is norm dense in X^*.

Definition 1. *We recall from* [4] *that x^* is said to be a point of norm weak usc for ρ, if given a weak neighbourhood V of 0 in X, there is a $\delta > 0$ such that for a norm attaining x_1^* with $\|x^* - x_1^*\| < \delta$ implies $\rho(x_1^*) \subset V + \rho(x^*)$.*

Thus by Lemma 2.2 in [4] the denseness of the state space is equivalent to x being a point of norm-weak upper-semi-continuity of the preduality map $\rho(x) = S_x$ on X^{**}. More generally, the same conclusions apply for any point of norm-weak usc in a dual space X^* (it is assumed that the functional attains its norm).

Since in many classical situations, like in ℓ^p spaces $(1 < p < \infty)$ or when X, Y are reflexive with the metric approximation property, one has the inclusion map on the space of compact operators, $\mathcal{K}(X,Y) \subset \mathcal{L}(X,Y)$ is the canonical embedding of $\mathcal{K}(X,Y)$ in its bidual $\mathcal{L}(X,Y)$ (see the discussion on page 247 in [2]), we take the help of norm-weak usc operators in $\mathcal{K}(X,Y)$ or $\mathcal{L}(X,Y)$ to evaluate the limits. We refer to the monograph [2], for basic theory of tensor products and structure of spaces of operators.

We show that when the projective tensor product space $X \otimes_\pi Y$ has the Radon-Nikodym property (RNP), then for $A \in \mathcal{L}(X,Y^*)$, $\lim_{t\to 0^+}\frac{\|A+tB\|-\|A\|}{t} = sup\{x^{**}(B^*(y^*)) : x^{**} \otimes y^* \in S_A\}$, when A is a point of norm-weak usc for the preduality map on $\mathcal{L}(X,Y^*) = (X \otimes_\pi Y)^*$.

canonical projection Q on X^{***} defined above is a L-projection. See Chapter III of [5] for examples and Chapter VI when spaces of compact operators $\mathcal{K}(X,Y)$ is a M-ideal, when $\mathcal{L}(X^{**},Y^{**})$ is identified as its bidual. When X is a M-ideal in its bidual, X^* has the RNP. See [5] Theorem III.3.1.

It is easy to see that unitaries in a C^*-algebra A, remain as points of norm-weak usc for the preduality map in all duals of higher even order of A (as they are still unitaries in the bigger space). The following phenomenon exhibits weak*-dense set of points of norm-weak usc for the preduality map on X^{**}. We do not know how to determine the largeness of points of norm-weak usc for the preduality map on X^{**} in the category sense, when X is a M-ideal in its bidual.

Proposition 9. *Let X be a non-reflexive space that is a M-ideal in its bidual. a) Any $0 \neq x \in X$ is a point of norm-weak usc for the preduality map on X^{**} and it continues to be so in all higher duals of even order of X.*

b) If $\tau \in X^{**}$ is such that τ attains its norm on X^* and $d(\tau, X) < \|\tau\|$, then τ is a point of norm-weak usc for the preduality map on X^{**}. This behaviour continues in all the duals of even order of X.

Proof. (a): Since the projection Q is a L-projection, $X^{***} = X^* \bigoplus_1 X^{\perp}$, we see that x^{***} is the unique extension of its restriction to X and the kernel annihilates X. Hence the conclusion follows.

It was shown in [8] that under the hypothesis X continues to be a M-ideal of all the higher order even duals of X (all in canonical embeddings). Hence the conclusion follows from Proposition 6.

(b): Since $X^{***} = X^* \bigoplus_1 X^{\perp}$ for the canonical projection, the conclusion follows from arguments similar to the ones given during the proof of Theorem 5.

References

1. T. S. S. R. K. Rao, *Computing sub differential limits of operators on Banach spaces*, arXiv:2212.09650, 2022.
2. G. Godefroy and V. Indumathi, *Norm-to-weak upper semi-continuity of the duality and preduality mappings*, Set-Valued Analysis 10 (2002), 317–330.
3. P. Harmand, D. Werner, and W. Werner, *M-ideals in Banach spaces and Banach algebras*, Lecture Notes in Mathematics 1547, Springer, 1993.